# BABU BANARASI DAS UNIVERSITY



## SCHOOL OF COMPUTER APPLICATIONS

## PROJECT ON

## "MOVIE TICKET BOOKING SYSTEM USING MONGODB "

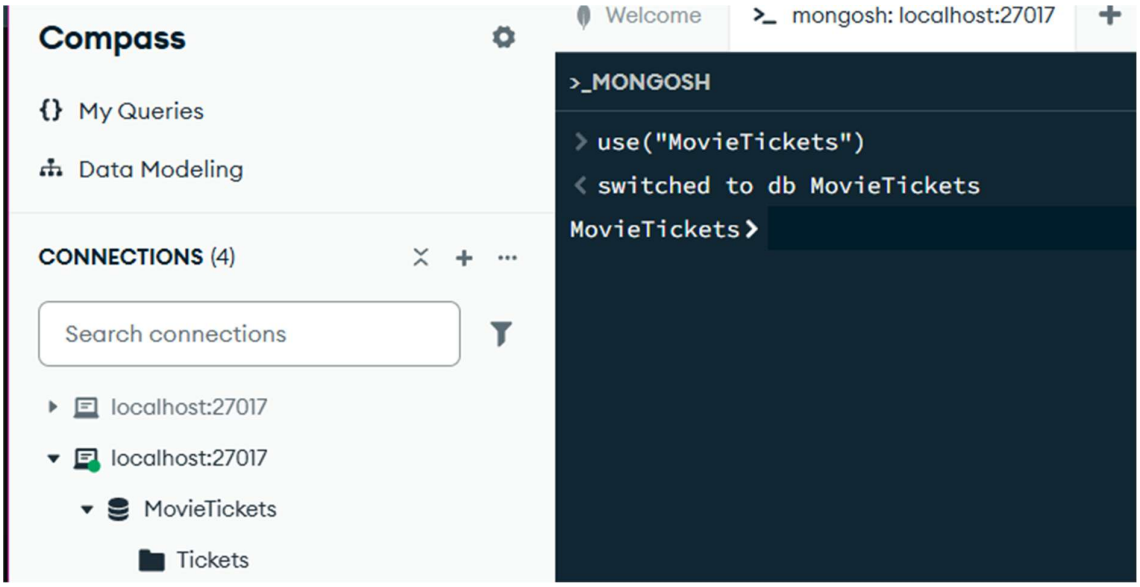

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**(2026-2027)**

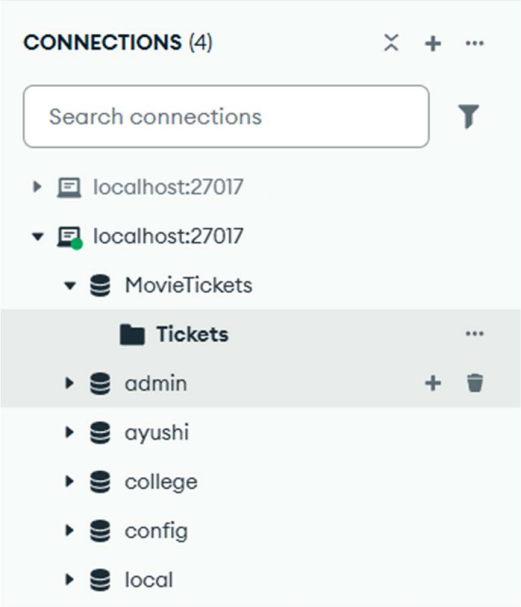
## CREATE DATABASE

**MongoDB Query:** use MovieTickets




## CREATE COLLECTION

**MongoDB QUERY:** db.createCollection("tickets")



# INSERT DATA

## MongoDB QUERY:

```
db.tickets.insertMany([

 {ticket_id:1, movie:"Pathaan", theatre:"PVR Lucknow", seats:2, price:400, city:"Lucknow", date:"2026-09-01", status:"Booked"},

 {ticket_id:2, movie:"KGF Chapter 2", theatre:"INOX Delhi", seats:3, price:350, city:"Delhi", date:"2026-09-02", status:"Booked"},

 {ticket_id:3, movie:"RRR", theatre:"Cinepolis Mumbai", seats:1, price:500, city:"Mumbai", date:"2026-09-03", status:"Cancelled"},

 {ticket_id:4, movie:"Jawan", theatre:"PVR Bangalore", seats:4, price:600, city:"Bangalore", date:"2026-09-04", status:"Booked"},

 {ticket_id:5, movie:"Gadar 2", theatre:"INOX Lucknow", seats:2, price:300, city:"Lucknow", date:"2026-09-05", status:"Booked"},

 {ticket_id:6, movie:"Animal", theatre:"PVR Delhi", seats:3, price:450, city:"Delhi", date:"2026-09-06", status:"Cancelled"},

 {ticket_id:7, movie:"Pathaan", theatre:"Cinepolis Pune", seats:2, price:400, city:"Pune", date:"2026-09-07", status:"Booked"},

 {ticket_id:8, movie:"Dunki", theatre:"PVR Hyderabad", seats:1, price:350, city:"Hyderabad", date:"2026-09-08", status:"Booked"},

 {ticket_id:9, movie:"Tiger 3", theatre:"INOX Chennai", seats:5, price:550, city:"Chennai", date:"2026-09-09", status:"Booked"},

 {ticket_id:10, movie:"KGF Chapter 2", theatre:"PVR Lucknow", seats:2, price:350, city:"Lucknow", date:"2026-09-10", status:"Cancelled"}

])
```

```
> db.tickets.insertMany([
   {ticket_id:1, movie:"Pathaan", theatre:"PVR Lucknow", seats:2, price:400, city:"Lucknow", date:"2026-09-01", status:"Booked"},
   {ticket_id:2, movie:"KGF Chapter 2", theatre:"INOX Delhi", seats:3, price:350, city:"Delhi", date:"2026-09-02", status:"Booked"},
   {ticket_id:3, movie:"RRR", theatre:"Cinepolis Mumbai", seats:1, price:500, city:"Mumbai", date:"2026-09-03", status:"Cancelled"},
   {ticket_id:4, movie:"Jawan", theatre:"PVR Bangalore", seats:4, price:600, city:"Bangalore", date:"2026-09-04", status:"Booked"},
   {ticket_id:5, movie:"Gadar 2", theatre:"INOX Lucknow", seats:2, price:300, city:"Lucknow", date:"2026-09-05", status:"Booked"},
   {ticket_id:6, movie:"Animal", theatre:"PVR Delhi", seats:3, price:450, city:"Delhi", date:"2026-09-06", status:"Cancelled"},
   {ticket_id:7, movie:"Pathaan", theatre:"Cinepolis Pune", seats:2, price:400, city:"Pune", date:"2026-09-07", status:"Booked"},
   {ticket_id:8, movie:"Dunki", theatre:"PVR Hyderabad", seats:1, price:350, city:"Hyderabad", date:"2026-09-08", status:"Booked"},
   {ticket_id:9, movie:"Tiger 3", theatre:"INOX Chennai", seats:5, price:550, city:"Chennai", date:"2026-09-09", status:"Booked"},
   {ticket_id:10, movie:"KGF Chapter 2", theatre:"PVR Lucknow", seats:2, price:350, city:"Lucknow", date:"2026-09-10", status:"Cancel
 ])
< {
   acknowledged: true,
   insertedIds: {
     '0': ObjectId('6aba11b1fe612c60921d774c'),
     '1': ObjectId('6aba11b1fe612c60921d774d'),
     '2': ObjectId('6aba11b1fe612c60921d774e'),
     '3': ObjectId('6aba11b1fe612c60921d774f'),
     '4': ObjectId('6aba11b1fe612c60921d7750'),
     '5': ObjectId('6aba11b1fe612c60921d7751'),
     '6': ObjectId('6aba11b1fe612c60921d7752'),
     '7': ObjectId('6aba11b1fe612c60921d7753'),
     '8': ObjectId('6aba11b1fe612c60921d7754'),
     '9': ObjectId('6aba11b1fe612c60921d7755')
   }
 }
```

# MongoDB QUERIES/OPERATIONS

## DISPLAY ALL TICKETS

**MongoDB Query:** db. tickets.find()

```
> db.tickets.find()
< {
   _id: ObjectId('6aba11b1fe612c60921d774c'),
   ticket_id: 1,
   movie: 'Pathaan',
   theatre: 'PVR Lucknow',
   seats: 2,
   price: 400,
   city: 'Lucknow',
   date: '2026-09-01',
   status: 'Booked'
 }
 {
   _id: ObjectId('6aba11b1fe612c60921d774d'),
   ticket_id: 2,
   movie: 'KGF Chapter 2',
   theatre: 'INOX Delhi',
   seats: 3,
   price: 350,
   city: 'Delhi',
   date: '2026-09-02',
   status: 'Booked'
 }
 {
   _id: ObjectId('6aba11b1fe612c60921d774e'),
   ticket_id: 3,
   movie: 'RRR',
   theatre: 'Cinepolis Mumbai',
```

## $eq operator — Equal To!

**MongoDB QUERY:** db.tickets.find({price: {$eq: 400}})

```
> db.tickets.find({price: {$eq: 400}})
< {
    _id: ObjectId('6aba11b1fe612c60921d774c'),
    ticket_id: 1,
    movie: 'Pathaan',
    theatre: 'PVR Lucknow',
    seats: 2,
    price: 400,
    city: 'Lucknow',
    date: '2026-09-01',
    status: 'Booked'
  }
  {
    _id: ObjectId('6aba11b1fe612c60921d7752'),
    ticket_id: 7,
    movie: 'Pathaan',
    theatre: 'Cinepolis Pune',
    seats: 2,
    price: 400,
    city: 'Pune',
    date: '2026-09-07',
    status: 'Booked'
  }
```

## $gt — Greater Than

**MongoDB QUERY:** db. tickets.find({price: {$gt: 450}})

```
> db.tickets.find({price: {$gt: 450}})
< {
    _id: ObjectId('6aba11b1fe612c60921d774e'),
    ticket_id: 3,
    movie: 'RRR',
    theatre: 'Cinepolis Mumbai',
    seats: 1,
    price: 500,
    city: 'Mumbai',
    date: '2026-09-03',
    status: 'Cancelled'
  }
  {
    _id: ObjectId('6aba11b1fe612c60921d774f'),
    ticket_id: 4,
    movie: 'Jawan',
    theatre: 'PVR Bangalore',
    seats: 4,
    price: 600,
    city: 'Bangalore',
    date: '2026-09-04',
    status: 'Booked'
  }
```

## $lt — Less Than

**MongoDB QUERY:** db. tickets.find({price: {$lt: 350}})

```
> db.tickets.find({price: {$lt: 350}})
< {
    _id: ObjectId('6aba11b1fe612c60921d7750'),
    ticket_id: 5,
    movie: 'Gadar 2',
    theatre: 'INOX Lucknow',
    seats: 2,
    price: 300,
    city: 'Lucknow',
    date: '2026-09-05',
    status: 'Booked'
  }
```

## $gte — Greater Than or Equal To

**Mongodb QUERY:** db.tickets.find({price: {$gte: 550}})

```
> db.tickets.find({price: {$gte: 550}})
< {
    _id: ObjectId('6aba11b1fe612c60921d774f'),
    ticket_id: 4,
    movie: 'Jawan',
    theatre: 'PVR Bangalore',
    seats: 4,
    price: 600,
    city: 'Bangalore',
    date: '2026-09-04',
    status: 'Booked'
  }
  {
    _id: ObjectId('6aba11b1fe612c60921d7754'),
    ticket_id: 9,
    movie: 'Tiger 3',
    theatre: 'INOX Chennai',
    seats: 5,
    price: 550,
    city: 'Chennai',
    date: '2026-09-09',
    status: 'Booked'
```

## $lte — Less Than or Equal To

**MongoDB QUERY:** db. tickets.find({price: {$lte: 300}})

```
> db.tickets.find({price: {$lte: 300}})
< {
    _id: ObjectId('6aba11b1fe612c60921d7750'),
    ticket_id: 5,
    movie: 'Gadar 2',
    theatre: 'INOX Lucknow',
    seats: 2,
    price: 300,
    city: 'Lucknow',
    date: '2026-09-05',
    status: 'Booked'
  }
```

## $and — AND Operator

### MongoDB QUERY:

db.tickets.find({$and:[{price:{$gt:400}},{status:"Booked"}]})

```
> db.tickets.find({$and:[{price:{$gt:400}},{status:"Booked"}]})
< {
    _id: ObjectId('6aba11b1fe612c60921d774f'),
    ticket_id: 4,
    movie: 'Jawan',
    theatre: 'PVR Bangalore',
    seats: 4,
    price: 600,
    city: 'Bangalore',
    date: '2026-09-04',
    status: 'Booked'
  }
  {
    _id: ObjectId('6aba11b1fe612c60921d7754'),
    ticket_id: 9,
    movie: 'Tiger 3',
    theatre: 'INOX Chennai',
    seats: 5,
    price: 550,
    city: 'Chennai',
    date: '2026-09-09',
    status: 'Booked'
  }
```

## $or — OR Operator

### MongoDB QUERY:

db. tickets.find({$or:[{city:"Lucknow"},{city:"Delhi"}]})



```
> db.tickets.find({$or:[{city:"Lucknow"},{city:"Delhi"}]})
< {
    _id: ObjectId('6aba11b1fe612c60921d774c'),
    ticket_id: 1,
    movie: 'Pathaan',
    theatre: 'PVR Lucknow',
    seats: 2,
    price: 400,
    city: 'Lucknow',
    date: '2026-09-01',
    status: 'Booked'
  }
  {
    _id: ObjectId('6aba11b1fe612c60921d774d'),
    ticket_id: 2,
    movie: 'KGF Chapter 2',
    theatre: 'INOX Delhi',
    seats: 3,
    price: 350,
    city: 'Delhi',
    date: '2026-09-02',
    status: 'Booked'
  }
```

## Sort — Ascending

**MongoDB QUERY:** db.tickets.find().sort({price:1})

```
> db.tickets.find().sort({price:1})
< {
    _id: ObjectId('6aba11b1fe612c60921d7750'),
    ticket_id: 5,
    movie: 'Gadar 2',
    theatre: 'INOX Lucknow',
    seats: 2,
    price: 300,
    city: 'Lucknow',
    date: '2026-09-05',
    status: 'Booked'
  }
  {
    _id: ObjectId('6aba11b1fe612c60921d774d'),
    ticket_id: 2,
    movie: 'KGF Chapter 2',
    theatre: 'INOX Delhi',
    seats: 3,
    price: 350,
    city: 'Delhi',
    date: '2026-09-02',
    status: 'Booked'
  }
```

## Sort — Descending

**MongoDB QUERY:** db.tickets.find().sort({price:-1})

```
< {
    _id: ObjectId('6aba11b1fe612c60921d774f'),
    ticket_id: 4,
    movie: 'Jawan',
    theatre: 'PVR Bangalore',
    seats: 4,
    price: 600,
    city: 'Bangalore',
    date: '2026-09-04',
    status: 'Booked'
  }
  {
    _id: ObjectId('6aba11b1fe612c60921d7754'),
    ticket_id: 9,
    movie: 'Tiger 3',
    theatre: 'INOX Chennai',
    seats: 5,
    price: 550,
    city: 'Chennai',
    date: '2026-09-09',
    status: 'Booked'
  }
```

## Limit

**MongoDB QUERY:** db.tickets.find().limit(5)

```
> db.tickets.find().limit(5)
< {
    _id: ObjectId('6aba11b1fe612c60921d774c'),
    ticket_id: 1,
    movie: 'Pathaan',
    theatre: 'PVR Lucknow',
    seats: 2,
    price: 400,
    city: 'Lucknow',
    date: '2026-09-01',
    status: 'Booked'
  }
  {
    _id: ObjectId('6aba11b1fe612c60921d774d'),
    ticket_id: 2,
    movie: 'KGF Chapter 2',
    theatre: 'INOX Delhi',
    seats: 3,
    price: 350,
    city: 'Delhi',
    date: '2026-09-02',
    status: 'Booked'
  }
  {
    _id: ObjectId('6aba11b1fe612c60921d774e'),
    ticket_id: 3,
    movie: 'RRR',
    theatre: 'Cinepolis Mumbai',
```

## Update

**MongoDB QUERY:**

db.tickets.updateOne({ticket_id:1},{$set:{price:500}})

```
> db.tickets.updateOne({ticket_id:1},{$set:{price:500}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

## Delete One

**MongoDB QUERY:** db. tickets.deleteOne({ticket_id:50})

```
> db.tickets.deleteOne({ticket_id:50})
< {
    acknowledged: true,
    deletedCount: 0
  }
```

## Search by City

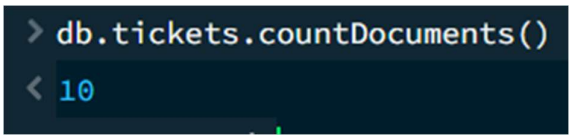
**MongoDB QUERY:** db. tickets.find({city:"Lucknow"})



```
> db.tickets.find({city:"Lucknow"})
< {
    _id: ObjectId('6aba11b1fe612c60921d774c'),
    ticket_id: 1,
    movie: 'Pathaan',
    theatre: 'PVR Lucknow',
    seats: 2,
    price: 500,
    city: 'Lucknow',
    date: '2026-09-01',
    status: 'Booked'
  }
  {
    _id: ObjectId('6aba11b1fe612c60921d7750'),
    ticket_id: 5,
    movie: 'Gadar 2',
    theatre: 'INOX Lucknow',
    seats: 2,
    price: 300,
    city: 'Lucknow',
    date: '2026-09-05',
    status: 'Booked'
  }
```

## Count Documents

**MongoDB QUERY:** db. tickets.countDocuments()



```
> db.tickets.countDocuments()
< 10
```

# MongoDB Operators Used

| Operator | Meaning | Example |
|---|---|---|
| 1. $eq | Equal to | price: {$eq:400} |
| 2. $gt | Greater than | price: {$gt:450} |
| 3. $lt | Less than | price: {$lt:350} |
| 4. $gte | Greater than or equal to | price: {$gte:550} |
| 5. $lte | Less than or equal to | price: {$lte:300} |
| 6. $and | All conditions true | $and:[...] |
| 7. $or | At least one condition is true | $or:[...] |

# CONCLUSION

In this project, a Movie Ticket Booking System was successfully created using MongoDB. A database named MovieTickets was created with a collection of 50 ticket records. Various MongoDB Query Operators such as $eq, $gt, $lt, $gte, $lte, $and, and $or were applied to filter and retrieve data efficiently. Update, Delete, Sort, Limit, and Count operations were also performed successfully.

Through this project, we learned how MongoDB operators work in real-world database applications and how NoSQL databases provide a flexible and efficient way to manage data.

# LEARNING OUTCOMES

- Learned how to create a MongoDB database and collection.
- Understood how to insert, display, update, and delete documents.
- Applied various MongoDB Query Operators to filter data.
- Learned how to sort, limit, and count documents in MongoDB.
- Understood the difference between SQL and NoSQL databases.

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*THANKYOU!*

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